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Roll no.:

MID-SEMESTER EXAMINATION, FEBRUARY-MARCH, 2025
4th Semester (Backlog)

Course Code: CMMTC13

Course Title: Applied Linear Algebra

Time: 1.5 Hours

Max. Marks: 25

Note: Attempt all questions in given order only. Missing data/information (if any), may be suitably assumed and mentioned in the answer.

Q. No.	Question	Marks	CO
1(a)	Check whether the set of positive rationals with $a * b = ab/2$ is a group or not, where $a, b \in \mathbb{Q}^+$. Justify your answer.	2.5	CO1
1(b)	Show that the set of vectors $(1, -2, 2), (3, -4, -1), (1, -1, 7)$, and $(5, 0, 0)$ do not span $\mathbb{R}^3$.	2.5	CO1
2(a)	Define subspace of a vector space. Show that the set $V = \{(x, y) \in \mathbb{R}^2 \mid xy \geq 0\}$ is not a vector subspace of $\mathbb{R}^2$.	2.5	CO1
2(b)	Show that $W_1 \oplus W_2 = \mathbb{R}^2$, where $W_1 = \{(x, 2x) : x \in \mathbb{R}\}$ and $W_2 = \{(x, 3x) : x \in \mathbb{R}\}$.	2.5	CO1
3(a)	Let $A = \{(1, 1, 2, 4), (2, -1, -5, 2), (1, -1, -4, 0), (2, 1, 1, 6)\}$. Prove or disprove the set A is linearly independent in $\mathbb{R}^4$.	2.5	CO1
3(b)	Determine whether the vectors $v_1 = (1, -3, 2)$, $v_2 = (2, 4, 1)$, $v_3 = (1, 1, 1)$ form a basis of $\mathbb{R}^3$. Justify your answer.	2.5	CO2
4(a)	Verify rank-nullity theorem for the linear transformation $T : P_2(\mathbb{R}) \rightarrow P_3(\mathbb{R})$ such that $T(f(x)) = xf(x) + f'(x)$.	2.5	CO2
4(b)	Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation defined by $T(x, y, z) = (x + 2y - 3z, 2x + 5y - 4z, x + 4y + z)$. Find the basis of image of T .	2.5	CO2
5(a)	Show that the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $T(x, y) = (x + y, x)$, for all $x, y \in \mathbb{R}$ is invertible.	2.5	CO2
5(b)	Show that the function $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by $T(x, y, z) = (2x + y + z, x + z + 1)$ is not a linear transformation.	2.5	CO2